

22 Lagrangians and torque, cont

Review

Torque-Free Rotation

$$\begin{aligned}\mathcal{L} = T_{\text{rot}} &= \frac{1}{2} \lambda (\dot{\theta} + \sin^2 \theta \dot{\phi}) + \frac{1}{2} \lambda_3 (\dot{\psi} + \dot{\phi} \cos \theta)^2 \\ P_{\psi} &= \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = \lambda_3 (\dot{\psi} + \dot{\phi} \cos \theta) \\ P_{\phi} &= \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \lambda \dot{\phi} \sin^2 \theta + \lambda_3 (\dot{\psi} + \dot{\phi} \cos \theta) \cos \theta \\ &= \lambda \dot{\phi} \sin^2 \theta + P_{\psi} \cos \theta\end{aligned}$$

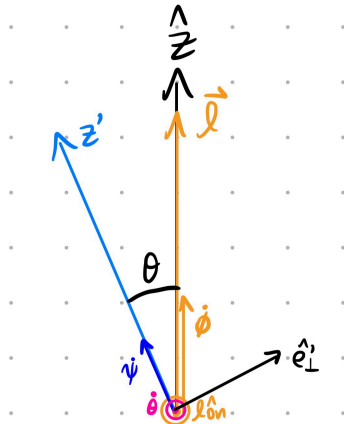
Today we will solve for the motion using a Lagrangian.

$\dot{\phi}$: Precession

$\dot{\psi}$: Spin, about the $\hat{z}$ axis.

$\dot{\theta}$: Nutation. Along the line of nodes.

If we choose $\hat{z}$ along the $\vec{l}$ (angular momentum direction), we can build up our coordinate frame.



We can write out the angular momentum in this new frame. Lets also take the assumption that this body is Axisymmetric ($\lambda_1 = \lambda_2 = \lambda$)

$$\begin{aligned}l_{z'} &= l \cos \theta &= \lambda_3 \omega_{z'} &= \lambda_3 (\dot{\psi} + \dot{\phi} \cos \theta) \\ l_{\perp'} &= l \sin \theta &= \lambda \omega_{\perp'} &= \lambda \dot{\phi} \sin \theta\end{aligned}$$

We get from here,

$$\dot{\phi} = \frac{l}{\lambda}$$

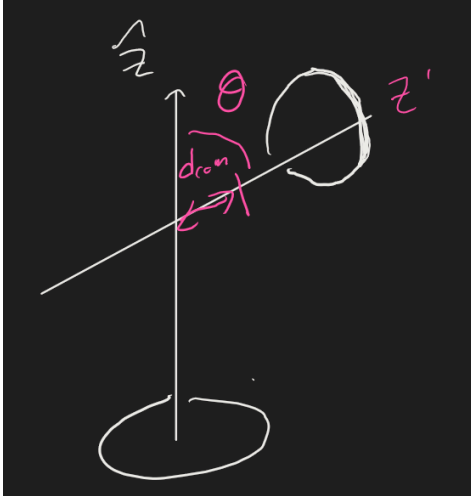
$$\begin{aligned}P_{\phi} &= \underbrace{(\lambda \dot{\phi} \sin \theta)}_{l_{\perp'}} \sin \theta + \underbrace{p_{\psi}}_{l_{z'}} \cos \theta \\ P_{\phi} &= l \sin^2 \theta + l \cos^2 \theta = l\end{aligned}$$

We can write out our Euler Lagrange equation:

$$\begin{aligned}\lambda \ddot{\theta} &= \lambda \dot{\phi}^2 \sin \theta \cos \theta - \lambda_3 (\dot{\psi} + \dot{\phi} \cos \theta) \dot{\phi} \sin \theta \\ &= \dot{\phi} \sin \theta (\lambda \dot{\phi} \cos \theta - P_{\psi}) \\ &= \dot{\phi} \sin \theta \left(\frac{\lambda l}{\lambda} \cos \theta - l \cos \theta \right) \\ \ddot{\theta} &= 0 \text{ If } \hat{z} \text{ along } \vec{l}\end{aligned}$$

Gyroscopes

In trying to model torque free procession, we made the gyroscope balanced with the center of mass on the hinge:



The center of mass is going to be moving around.

$$T = \frac{1}{2} M v_{\text{tagged point}}^2 + T_{\text{cross}} + T_{\text{rot}}$$

If we make the tagged point the hinge instead of the center of mass, then this will cancel out. Lets define $u = 0$ at the line of the fixed point so that the center of mass has a height $d \cos \theta$ above it.

$$T = T_{\text{rot}}^{\text{axisymmetric}} + M g d \cos \theta$$

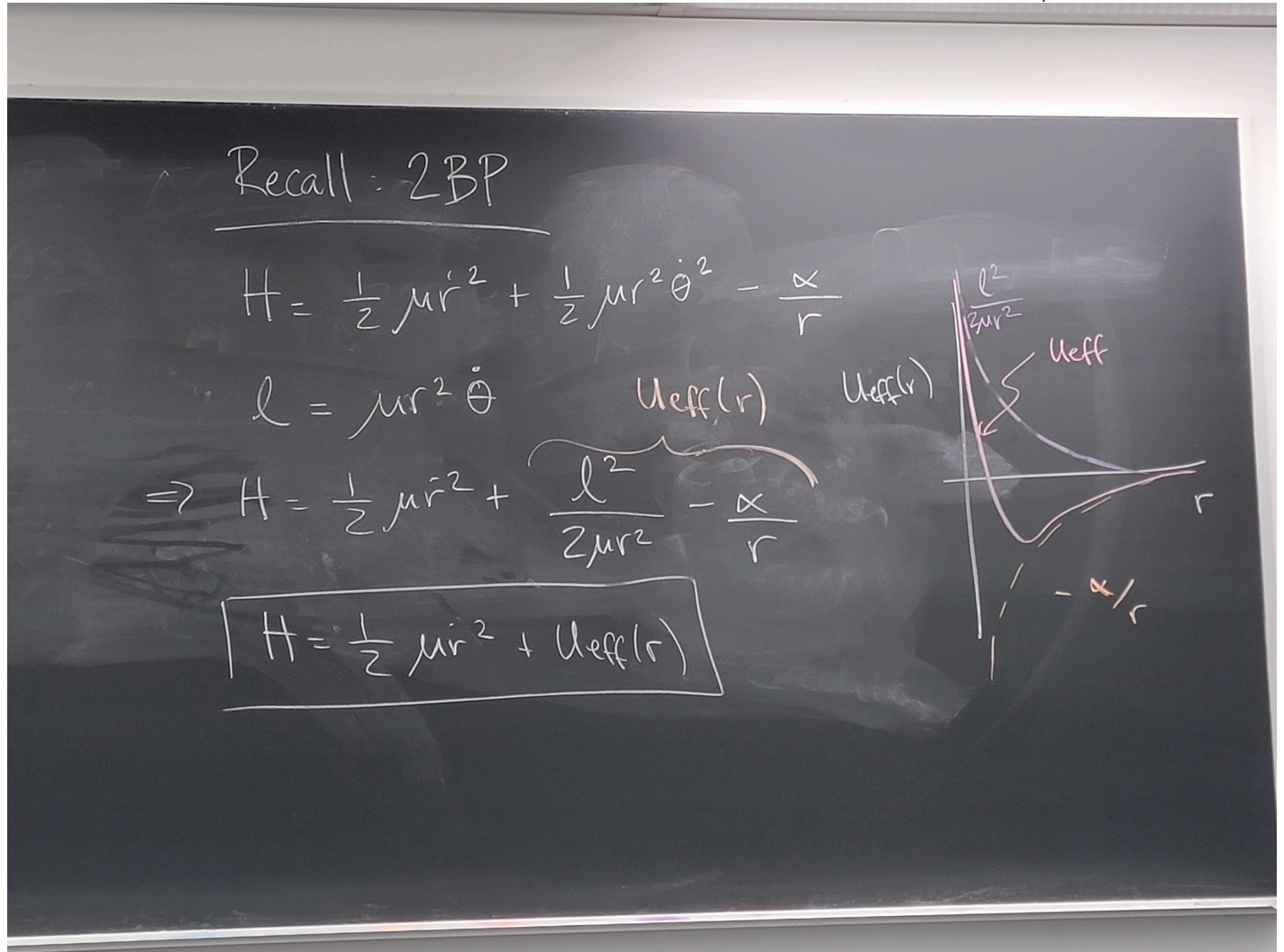
We get the Lagrangian:

$$\mathcal{L} = \frac{1}{2} \lambda (\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{1}{2} \lambda_3 (\dot{\psi} + \dot{\phi} \cos \theta)^2 - M g d \cos \theta$$

P_ψ and P_ϕ are both conserved, since ψ and ϕ don't show up in the Lagrangian. Because there is no time dependence and no time varying axes, the Hamiltonian $\mathcal{H}$ is conserved.

We can totally make this nice, its just the same as what we did with the two body problem.

For that, we reduced from 2 variables to 1 by taking advantage of conserved quantities (replacing $\dot{\theta}$ with $\frac{l}{\mu r^2}$).



Note that this $\dot{\phi}$ could be written as

$$\dot{\phi} = \frac{P_{\phi} - P_{\psi} \cos \theta}{\lambda \sin^2 \theta}$$

In our case now, the Hamiltonian is

$$\begin{aligned} \mathcal{H} &= T + U \\ &= \frac{1}{2} \lambda (\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{1}{2} \lambda_3 (\dot{\psi} + \dot{\phi} \cos \theta)^2 + Mgd \cos \theta \\ &= \frac{1}{2} \lambda \dot{\theta}^2 + \frac{P_{\psi}^2}{2\lambda_3} + \frac{(P_{\phi} - P_{\psi} \cos \theta)^2}{2\lambda \sin^2 \theta} + Mgd \cos \theta \end{aligned}$$

Note that $\mathcal{H}$ is constant, and that we just have terms that are either constant or only in terms of θ .

$$\begin{aligned} \frac{1}{2} \lambda \left(\frac{d\theta}{dt} \right)^2 &= H - u_{eff}(\theta) \\ d\theta &= \pm \sqrt{\frac{2}{\lambda} (H - U_{eff}(\theta))} dt \end{aligned}$$

This doesn't have an analytic solution, but qualitatively it looks like the 3 body problem still.

Effective Potential

θ is from 0 to π . Our U_{eff} will explode with θ is 0 or π , because we have a $\frac{1}{\sin^2 \theta}$.

Lets plot.

